

PRG200: Week 5 Deliverables

Team NASA

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PRG 200: Programming in the Cloud

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Introduction

The AI-Powered Virtual Fashion Assistant project aims to provide personalized fashion recommendations by analyzing a user's facial image and identifying their skin tone. The system uses Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision techniques to recommend suitable clothing colors, fashion styles, and personalized fashion moodboards. During Week 5, the team focused on researching core technologies, studying skin tone analysis methods, finalizing project requirements, establishing the development environment, and collecting datasets required for future implementation.

Research on Artificial Intelligence, Machine Learning, and Computer Vision

Artificial Intelligence (AI)

Artificial Intelligence refers to the development of computer systems capable of performing tasks that typically require human intelligence, such as decision-making, pattern recognition, and prediction. In this project, AI will be used to generate personalized recommendations based on the user's skin tone classification and fashion preferences.

Applications in the Project

- Personalized fashion recommendations
- Automated color palette selection
- Outfit style suggestions
- Moodboard generation

Machine Learning (ML)

Machine Learning is a subset of AI that allows systems to learn patterns from data and make predictions without being explicitly programmed. The project will utilize machine learning techniques to classify skin tones and recommend appropriate fashion choices.

Planned ML Tasks

- Skin tone classification
- Seasonal color palette prediction
- Recommendation generation
- Model evaluation and optimization

Computer Vision

Computer Vision enables computers to understand and process digital images. Since the project relies on user-uploaded selfies, Computer Vision is a critical component.

- Planned Computer Vision Tasks
- Face detection
- Facial image processing
- Skin region extraction
- Color analysis
- Feature extraction

Research Findings

After reviewing multiple AI and Computer Vision resources, the team identified OpenCV as the most suitable library for face detection and image processing. Python was selected as the

primary programming language due to its extensive support for machine learning and computer vision applications.

3. Study of Skin Tone Analysis Techniques

Skin tone analysis is one of the most important components of the project because all recommendations depend on accurately identifying the user's skin tone.

Skin Tone Categories

The team finalized the following skin tone categories:

- Fair
- Light
- Medium
- Olive
- Deep

These categories align with the project's proposed classification system and will serve as the basis for recommendation generation.

4. Finalized Project Requirements

Based on the project proposal and research findings, the following functional requirements were finalized.

Technical Requirements

The system must:

- Allow users to upload a facial image.

- Detect the face automatically.
- Extract skin regions from the image.
- Analyze skin tone characteristics.
- Classify the user into a skin tone category.
- Recommend seasonal color palettes.
- Suggest clothing colors and fashion styles.
- Generate a personalized fashion moodboard.
- Display recommendations through an interactive user interface.

Non-Technical Requirements

- User-friendly interface
- Accurate skin tone detection
- Fast processing time
- Scalable design
- Maintainable code structure

System Design Overview

Input

The user uploads a selfie image through the application.

Processing Steps

- Face Detection
- Skin Region Extraction
- Skin Tone Analysis

- Skin Tone Classification
- Color Palette Prediction
- Fashion Recommendation Generation
- Moodboard Creation

Output

The system will display:

- Detected skin tone category
- Recommended seasonal palette
- Suggested clothing colors
- Fashion style recommendations
- Personalized moodboard

Development Environment Setup

The team successfully configured the development environment required for implementation.

Software Installed

- Python
- Google Colab
- Visual Studio Code (VS Code)

Python Libraries Installed

- OpenCV
- NumPy
- Pandas

- Scikit-learn
- Matplotlib
- Pillow
- Streamlit

Environment Verification

All required tools were installed successfully and tested through sample scripts to verify compatibility and functionality.

7. Challenges Encountered

Several challenges were identified during Week 5:

- AI accuracy needs to be increased.
- The model classifies every skintone as “Dark”.

These challenges will be addressed during data preprocessing and model development in Week 6

Conclusion

Week 5 successfully established the foundation for the AI-Powered Virtual Fashion Assistant project. The team completed research on Artificial Intelligence, Machine Learning, Computer Vision, and skin tone analysis techniques. Project requirements and system design were finalized, the development environment was configured, and datasets were collected and organized. These accomplishments provide a strong foundation for Week 6, where dataset preprocessing, face detection implementation, skin tone classification, and recommendation system development will begin.